User Manual

FLIM Kit

FLK-01-1C1L



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03 Nov 2026

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1 General Information

This chapter provides essential information on the purpose of this manual, the manufacturer, regulatory compliance, warranty conditions, and the conventions used throughout this document.

1.1 About This Manual

This manual provides instructions for the installation, configuration, operation, maintenance, and disposal of the FLIM Kit. The product provides a compact instrument solution for single-photon time-resolved fluorescence lifetime spectroscopy and related laboratory applications.

The FLIM Kit combines a fiber-coupled picosecond pulsed laser source, a single-photon SPAD detector, a FLIM LABS data acquisition card, FLIM LABS software, optical fibers, and the required connection accessories.

Individual components included in the kit may be supplied with their own dedicated manuals containing additional safety, installation, and operating information. These manuals must also be read and followed before installation and operation.

This manual forms an integral part of the product and must remain accessible to the personnel responsible for the system. If the system is transferred to a third party, all accompanying documentation must be delivered with it.

This manual contains proprietary information and may not be reproduced or distributed, in whole or in part, without prior written consent of the manufacturer.

The components of the kit configurations are listed in Section 2.3. Depending on the ordered configuration, the laser wavelength and detector model may vary.



Warning Before performing any operation, the entire manual and the documentation of all supplied components must be read carefully and the relevant sections reviewed in detail.

1.1.1 User Qualification

The FLIM kit is intended for use by trained laboratory personnel familiar with laboratory electronic instrumentation and optical equipment. Personnel shall have received appropriate instruction regarding the installation, operation, maintenance, and safety precautions described in this manual.

The FLIM kit is not a toy and must not be used by children under 14 years of age. Persons with reduced physical, sensory, or cognitive abilities may operate the FLIM kit only under the supervision of qualified personnel.

1.1.2 Revision History

Document Revision: v1.1.1 - 03 Nov 2026

A certified copy of this manual is stored in the manufacturer's technical file for the legally required period. During this period, documentation may be requested from the manufacturer.

1.2 Symbols and Conventions

This manual includes two different levels of hazard warnings. The warnings may include various safety symbols. The hazard levels and safety symbols comply with the ISO Standard for Product Safety Signs and Labels.

Hazard Levels



Warning Indicates a hazardous situation that, if not avoided, could result in death or serious injury.



Caution Indicates a hazardous situation that, if not avoided, could result in minor or moderate injury.

Notice Indicates important information intended to prevent equipment damage, malfunction, or improper operation.

Safety Symbols

Table 1.1 shows a list of symbol types that can be found on the product or in the manual.

Table 1.1: Safety symbols and their meanings

Symbol	Form	Information
	Yellow triangle with a black border and a black symbol	Indicates a hazard.
	Circular sign with a red border, diagonal line, and black symbol	Identifies a prohibited action.
	Blue circle with a white symbol	Identifies a required action.

Warning Safety pictograms performing a protective function must not be removed, covered, or damaged. If a label becomes illegible or damaged, contact authorized service for replacement.

1.3 Manufacturer Information

Manufacturer: FLIM LABS S.r.l.
Address: Via della Farnesina, 3 00135 Rome (RM), Italy
Telephone: +39-329 761540
E-mail: info@flimlabs.com
Website: www.flimlabs.com

1.4 Regulatory Compliance

The FLIM LABS components supplied with the kit are individually CE-marked where applicable and are accompanied by their respective EU Declarations of Conformity in accordance with applicable European legislation.
The user is responsible for ensuring that the installed system complies with all applicable local regulations and is operated in accordance with the safety requirements and instructions provided in this manual and the accompanying component documentation.

1.5 Product and Component Labels

The components supplied with the product contain labels with regulatory markings and identification information including:

- Product name and model
- Manufacturer identification
- Regulatory markings where applicable
- Electrical specifications
- Serial number information

The product packaging may additionally contain kit-level identification information including the product number and configuration information.
Disposal requirements associated with WEEE markings are described in Section 7.2.

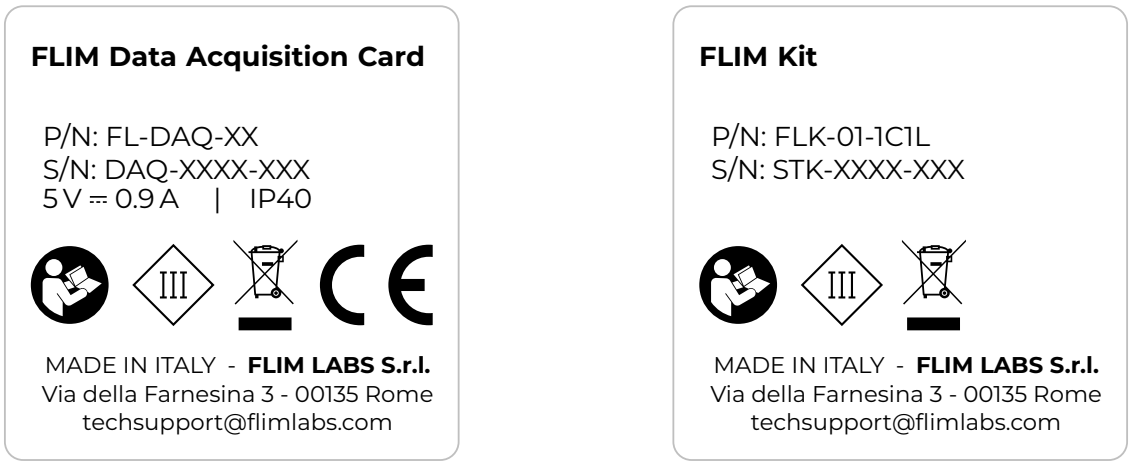


Figure 1.1: Product labels for the kit components

1.6 Warranty and Liability

Warranty

The manufacturer provides 24 months warranty for private customers and 12 months warranty for commercial customers. The warranty period begins on the date of delivery.

The warranty covers defects in materials and workmanship under normal and intended use. It is valid only if the product is installed, operated, and maintained according to this manual and the applicable manuals of the supplied components.

Unauthorized modification, improper use, failure to follow instructions, or repair by non-authorized personnel voids the warranty.

Limitation of Liability

The manufacturer shall not be liable for damages resulting from improper use, including but not limited to:

- Use outside the intended purpose
- Failure to follow this manual or applicable safety instructions
- Installation or operation by unqualified personnel
- Operation outside specified electrical, optical, or environmental limits
- Use of incompatible or non-compliant input signals or connected equipment
- Unauthorized modification, repair, or use of non-original spare parts
- Integration into systems not compliant with applicable safety regulations

2 System Description

This chapter provides an overview of the FLIM Kit, its intended application, major system components, and the overall system architecture.

The kit provides a compact platform for fluorescence lifetime spectroscopy, single-photon counting, TCSPC measurements, FLIM-Phasors analysis, image reconstruction, and fluorescence correlation spectroscopy.

2.1 Functional Description

The FLIM Kitenables time-resolved fluorescence lifetime measurements using pulsed optical excitation, single-photon detection, and FPGA-based TCSPC acquisition hardware.

The system operates by exciting the sample with a fiber-coupled picosecond pulsed laser source. Fluorescence photons emitted by the sample are detected by a fiber-coupled single-photon SPAD detector and converted into electrical timing signals. These signals are acquired by the FLIM LABS Data Acquisition Card and processed by FLIM LABS software for fluorescence lifetime analysis.

2.2 Intended Use

The FLIM Kitis intended for laboratory, research, education, and development applications involving fluorescence lifetime spectroscopy, single-photon time-resolved measurements, TCSPC, FLIM-Phasors, image reconstruction, and fluorescence correlation spectroscopy.

The kit is intended exclusively for indoor laboratory use under the environmental and operating conditions specified in this manual and in the manuals of the supplied components.

The kit is not intended for:

- Medical diagnosis or therapeutic applications
- Operation in explosive or hazardous environments
- Outdoor operation
- Safety-critical applications
- Use as a consumer product

Reasonably Foreseeable Misuse

Reasonably foreseeable misuse includes, but is not limited to:

- Modification of the supplied hardware or optical components
- Use of incompatible electrical or optical signals
- Bypassing laser safety precautions
- Operation outside specified environmental conditions
- Use without reading the supplied documentation
- Use by unqualified personnel
- Disconnecting optical fibers while the laser is active

2.3 System Components Overview

The standard FLIM Kitcontains the components listed in Table 2.1. The exact product numbers depend on the ordered configuration.

2.4 System Architecture

Figure 2.1 shows the overall architecture of the FLIM Kit.

The pulsed laser module provides the optical excitation signal. The laser output is routed through a fiber connection to the experimental optical setup or sample interface. Fluorescence photons are collected and routed to the SPAD detector.

The detector output and laser synchronization signal are connected to the FLIM LABS Data Acquisition Card. The DAQ records photon arrival timing information and transfers the acquired data to a connected computer through USB.

Table 2.1: System components overview

Component	FLIM LABS Part P/N	Description
Picosecond Pulsed Laser Module	PLD-XXX-SM-FC-U	Fiber-coupled excitation source for fluorescence lifetime measurements. Available wavelengths include 405, 445, 488, 520, 635, and 850 nm depending on the ordered configuration.
Single-Photon SPAD Detector	SPAD-050-C-X-FC	Fiber-coupled single-photon avalanche diode detector for fluorescence photon detection. Blue-shifted and red-shifted detector options are available.
FLIM Data Acquisition Card	FL-DAQ-0X	FPGA-based TCSPC acquisition hardware for photon timing measurements and time-resolved fluorescence analysis.
FLIM LABS Software	n.a.	Software for fluorescence lifetime acquisition, visualization, spectroscopy, FCS, and data analysis.
Optical fiber cables	n.a.	Fiber-optic cables used for laser delivery and fluorescence photon collection.
USB cables and power supplies	n.a.	Connection and power accessories required for operation of the supplied hardware components.

The acquired data are processed by FLIM LABS software for fluorescence lifetime analysis, spectroscopy, and related measurements.

2.5 System Requirements

The FLIM Data Acquisition Card requires a compatible Windows computer for configuration and operation.

- PC with multicore CPU
- 4 GB RAM
- USB interface for connecting the DAQ to the PC
- Windows 10 or later

Refer to the latest software documentation for updated system requirements.

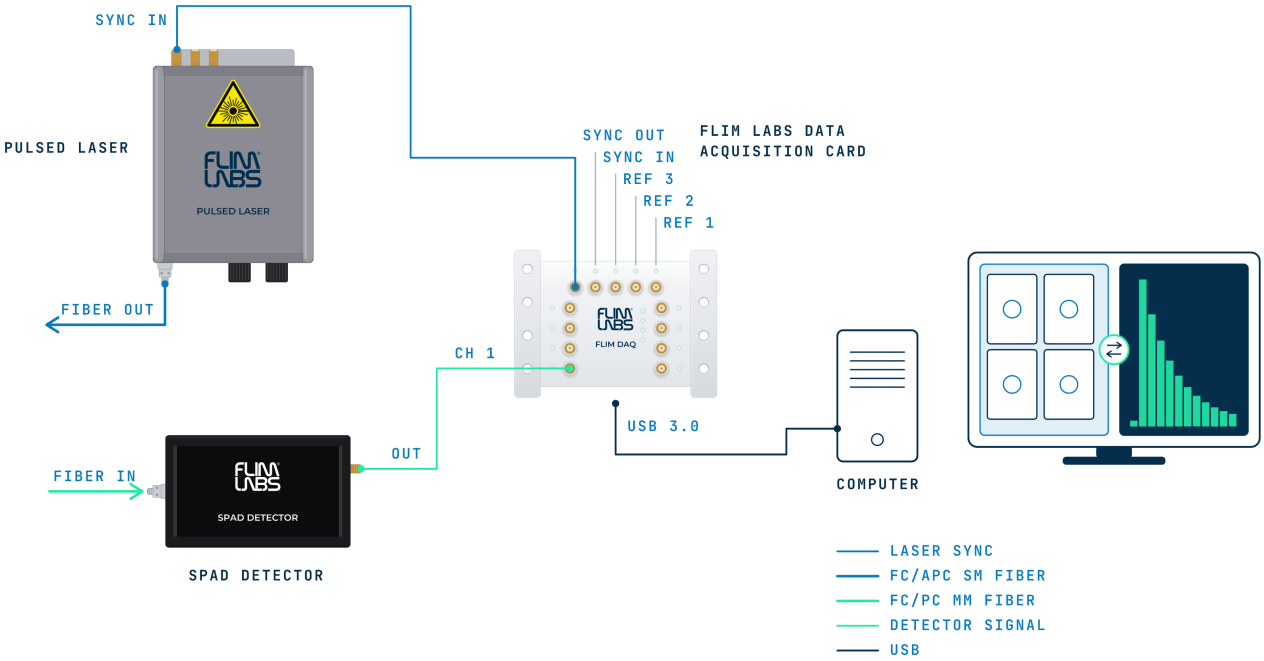


Figure 2.1: Connection diagram of the FLIM Kit.

3 Safety Information

3.1 General Safety Instructions

Before installation and operation, ensure that this manual and the documentation of all supplied components are complete and legible. If any part of the documentation is missing or unclear, contact the respective manufacturer before proceeding.

The product must be used only as described in this manual. Any operation not explicitly described in this manual is considered improper use.

This manual complements applicable national and local safety regulations. If local regulations impose stricter requirements, those regulations take precedence.

Despite the implemented safety measures, residual risks may remain if the system is not installed, operated, or maintained as specified.



Warning Before performing any installation or electrical connection procedure, disconnect power from all components unless explicitly instructed otherwise.

3.2 Laser Safety

The product may include a pulsed laser source or may be used together with laser sources.

Depending on the selected configuration, the system may emit visible and/or invisible laser radiation. Direct exposure to laser radiation may result in eye injury.

The laser source is delivered with its own dedicated product manual, safety labels, and regulatory information. These documents form part of the complete system documentation and must be read before installation and operation.



Caution Visible and invisible laser radiation.
Avoid direct eye exposure to laser radiation.
Do not disconnect optical fibers while the laser source is active.
Operate the system only under safe operating conditions and according to all supplied documentation.

3.3 Electrical Safety

The product contains electronic devices intended for laboratory use.

- Operate all components only with the specified power supplies and interfaces.
- Use only undamaged and properly rated cables.
- Disconnect power before modifying electrical or signal connections.
- Do not open the enclosures of electronic devices.
- There are no user-serviceable internal components.
- Keep all electronic components away from water sources and other liquids.
- Ensure that cables are routed to avoid mechanical damage and accidental disconnection.
- Unauthorized repair or modification is not permitted.

3.4 Optical Safety and Fiber Handling

Optical fibers and connectors must be handled carefully to avoid damage, contamination, or unintended exposure to optical radiation.

- Keep fiber connectors clean and protected with dust caps when not in use.
- Do not bend optical fibers below their specified minimum bending radius.
- Do not inspect active fiber outputs directly.
- Ensure that all fiber connections are secure before enabling the laser.

3.5 Environmental and Operating Conditions

The product is designed for indoor laboratory use only.

Operate the supplied components only within the environmental limits specified in the respective component manuals.

Typical operating conditions include:

- Indoor laboratory environment
- Stable, vibration-free installation surface
- Non-condensing humidity conditions
- Adequate ventilation around electronic components
- Protection from liquids, dust, and mechanical impact

Do not operate the system in explosive atmospheres or in the presence of flammable gases.

Ensure adequate ambient lighting during installation and adjustment procedures to avoid connection errors or accidental damage to optical components and connectors.

3.6 Modification and Improper Use

The supplied components must not be modified, altered, or repaired by unauthorized personnel.

4 Setup

This chapter describes the correct procedure for unpacking, installing, connecting, and verifying the FLIM Kitbefore operation.

4.1 Scope of Delivery

The exact scope of delivery depends on the ordered system configuration.

Table 4.1: Kit configuration and scope of delivery

Component	Quantity
Picosecond Pulsed Laser Module	1
Single-Photon SPAD Detector	1
FLIM Data Acquisition Card	1
FLIM LABS Software	1
Optical fiber cables	As configured
USB cables and power supplies	1 set

4.2 Unpacking and Inspection

After unpacking, carefully inspect all components and accessories:

- Check for visible damage.
- Check for dents, deformation, cracks, or signs of impact.
- Verify that all connectors and cables are intact.
- Inspect optical connectors and optical surfaces for contamination or damage.
- Verify that all ordered accessories are present.

If any damage, missing parts, or abnormalities are detected, stop the installation and contact the manufacturer before proceeding.



Warning Keep packaging materials out of reach of children and dispose of them according to local regulations.

4.3 Installation Requirements

The installation area must provide:

- Adequate workspace around the optical setup
- Stable laboratory surfaces
- Adequate ventilation for electronic components
- Protection from liquids and excessive dust
- Safe cable routing without tripping hazards
- Suitable laser safety measures for the selected configuration

4.4 Hardware Installation

Figure 2.1 shows an overview of all system components and their connections. Install the supplied components in locations that allow safe cable routing and stable operation.

4.5 Optical Connections

- Connect the pulsed laser module to the optical setup using the supplied single mode optical fiber.
- Connect the SPAD detector to the optical detection path using the supplied multimode optical fiber patch cable according to the experimental configuration.

4.6 Electrical Connections

All electrical connections must be performed by qualified personnel who have read this manual. Refer to the respective component manuals for connector locations.

Laser Trigger Connection

Connect the laser trigger signal from the laser *Sync-OUT* port to the *Sync-IN* port of the DAQ using an SMA cable.

Signal Input Connections

Connect the output signal of the SPAD detector to an available input channel on the DAQ.

USB Connection (Power and Data)

- Connect the DAQ to the host computer using the USB 3.0 micro-B cable. The USB connection provides both:
  - 5 V DC power supply to the device
  - Data transfer and communication with the host PC
- Connect the Laser to a power supply using a USB Type-C cable.
- Connect the SPAD to a power supply using a USB Type-C cable.

Connect all devices only to compatible power supplies and USB interfaces. Refer to the respective manuals for power ratings.

Do not apply power to the system until all optical and electrical connections have been completed and verified.

4.7 Example Imaging Configurations

Figures 4.1 and 4.1 show examples to demonstrate the integration of the FLIM Kit with microscope components and different detector configurations.

The actual configuration may vary depending on the microscope, detectors, and application requirements.

4.8 Software Installation and Initial Verification

Data acquisition and analysis are performed using FLIM LABS software.

After completing the hardware installation and software installation:

1. Verify that all optical and electrical connections are complete.
2. Power on the connected computer.
3. Power on all supplied hardware components.
4. Start the FLIM LABS software.
5. Verify that the FLIM Data Acquisition Card is detected.
6. Verify that detector and synchronization signals are present.
7. Configure the acquisition settings according to the experiment.
8. Perform a test acquisition before starting the final measurement.

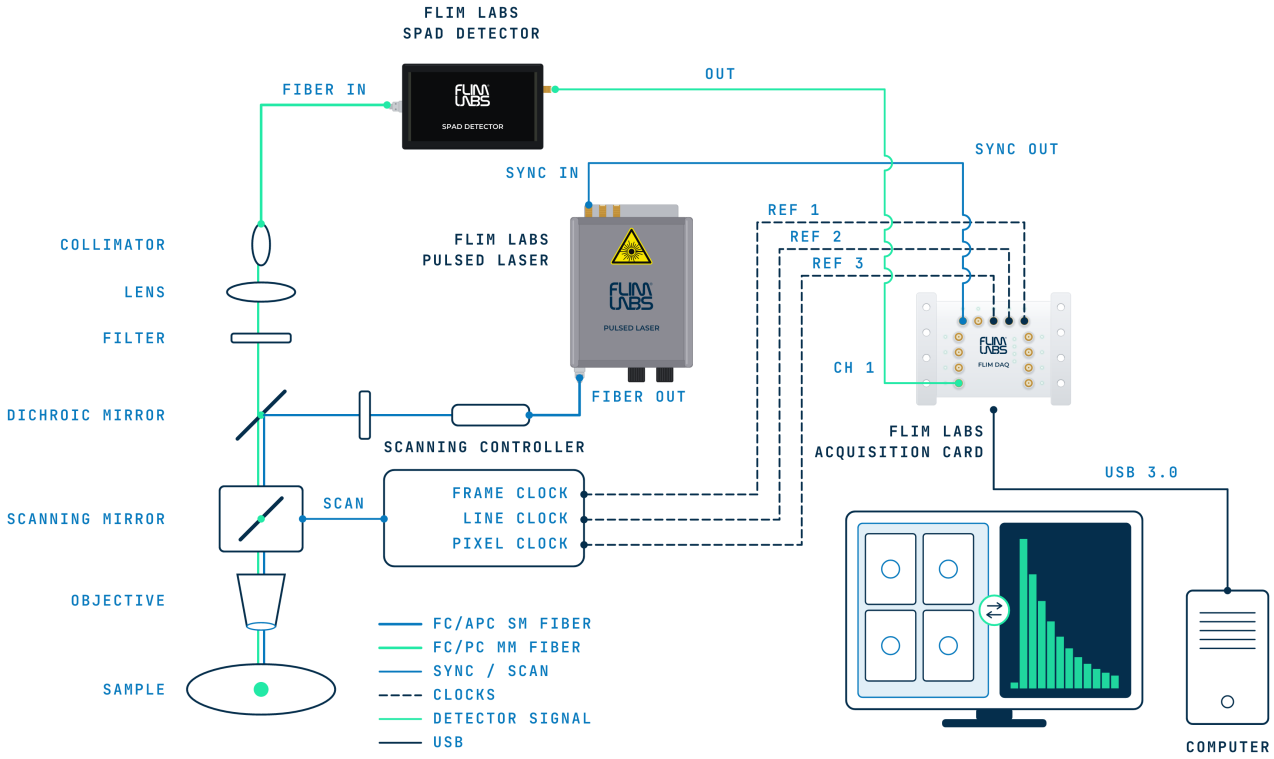


Figure 4.1: Example FLIM imaging setup illustrating the integration of the FLIM Kit with a typical fluorescence microscope.

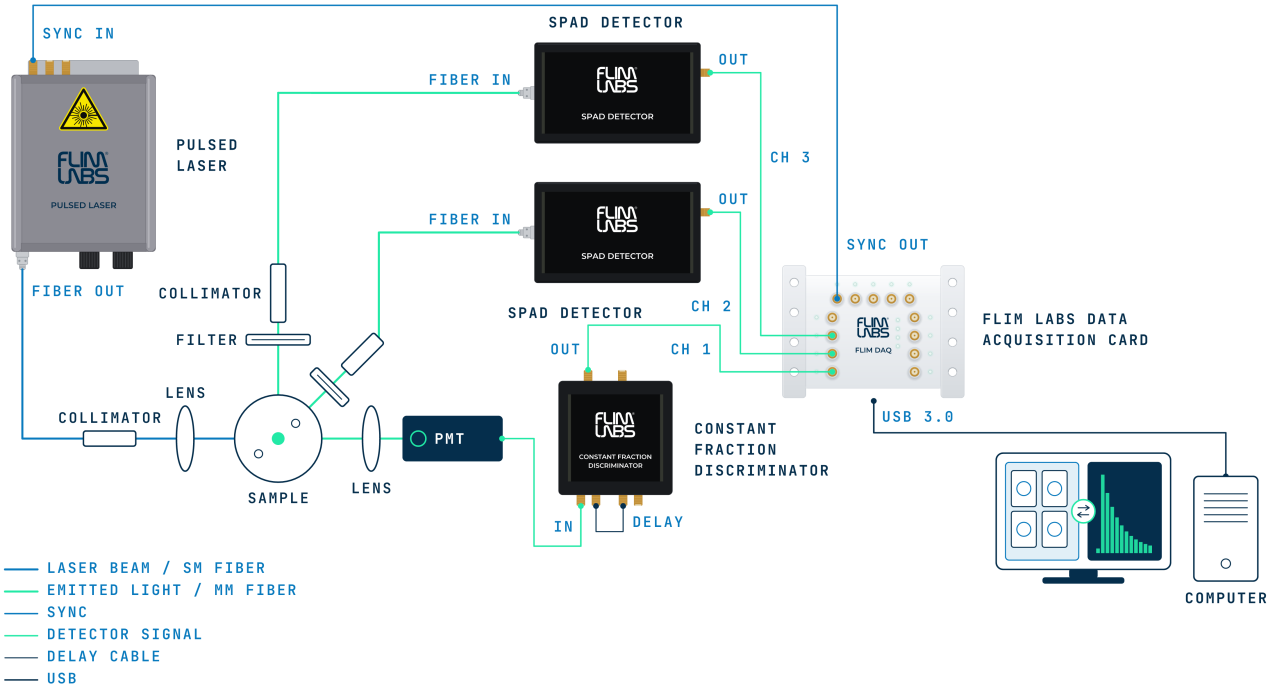


Figure 4.2: Example three-channel FLIM acquisition configuration using three detector channels for simultaneous fluorescence lifetime measurements.

5 Operation

This chapter describes the typical workflow for operating the FLIM Kit.

5.1 System Operation Principle

During operation, the pulsed laser generates optical excitation pulses. The detector records fluorescence photons generated by the sample. The FLIM Data Acquisition Card measures photon timing information relative to the selected timing reference.

The software displays, records, and processes the acquired fluorescence lifetime data.

5.2 Startup Procedure

1. Verify that all optical and electrical connections are complete.
2. Ensure that laser safety precautions are in place.
3. Power on the connected computer.
4. Power on all supplied hardware components.
5. Start the FLIM LABS software.
6. Load or create the required acquisition configuration.
7. Verify that the FLIM Data Acquisition Card is detected correctly.
8. Verify that detector and synchronization signals are present.
9. Start acquisition.

5.3 Acquisition Workflow

Typical measurements may include:

- Fluorescence lifetime spectroscopy
- Single-photon counting
- TCSPC acquisition
- FLIM-Phasors analysis
- Fluorescence correlation spectroscopy

5.4 Data Export

The supplied software supports export of acquired measurement data for further analysis and storage.

Users should verify the selected export location, file name, and acquisition settings before starting long measurements.



Notice Verify that data export is configured correctly before starting long acquisitions to avoid unintended data loss.

Detailed information about export formats and post-processing tools is provided in the dedicated software manual.

5.5 Shutdown Procedure

1. Stop acquisition in the software.
2. Save and export all required measurement data.
3. Disable the laser output.
4. Close the software.
5. Power down the supplied hardware components.
6. Disconnect power only after the system has been safely shut down.

5.6 Troubleshooting

Table 5.1: Common issues, possible causes, and solutions

Issue	Possible Cause	Solution
No hardware detected	USB connection or driver issue.	Check the USB connection, restart the software, and reinstall the driver if required.
No synchronization signal detected	Incorrect sync connection or trigger configuration.	Check the synchronization cable, signal source, and trigger settings.
No photon counts detected	Detector signal missing, laser inactive, or incorrect input selected.	Check detector output, laser status, input channel selection, and cable connection.
Unexpected count rate	Incorrect optical alignment, excessive background light, or detector saturation.	Check optical alignment, reduce background light, and adjust laser power or attenuation.
Unstable acquisition	USB bandwidth or connection issue.	Use a direct USB connection and avoid unsupported USB hubs.

6 Maintenance and Service

This chapter describes the cleaning, preventive maintenance, corrective maintenance, and authorized service procedures for the supplied hardware components. All maintenance activities must be performed by qualified personnel who have read and understood this manual.

Maintenance must be carried out in accordance with this manual and applicable workplace safety regulations. The maintenance area should provide adequate illumination; a minimum lighting level of 200 lux is recommended.



Before performing any cleaning or maintenance operation, switch off all powered components and disconnect them from their power sources.

6.1 Cleaning

The system components require periodic external cleaning to ensure reliable operation.

- Remove dust or dirt from exterior surfaces using a dry or slightly damp lint-free cloth.
- A mild, neutral detergent may be used if necessary.
- Do not allow liquids to enter connectors, optical interfaces, or electronic components.
- Avoid the use of solvents, aggressive chemicals, or abrasive materials.

For connectors and hard-to-reach areas, such as USB ports or ventilation openings, a soft-bristled brush or compressed air may be used.

Optical Port and Connector Cleaning

Proper cleaning of optical connectors is essential to ensure reliable optical coupling and to prevent signal attenuation or permanent damage to optical interfaces.

Always inspect the ferrule end faces of fiber patch cables and optical bulkhead connectors before making optical connections. Ferrule end faces may be cleaned using commercially available lint-free fiber cleaning wipes or by applying a small amount of isopropanol (IPA) to a clean lint-free cloth.

Optical bulkhead connectors are best cleaned using a dedicated fiber-optic cleaning pen or cleaning stick designed for the connector type in use. Follow the instructions provided by the manufacturer of the cleaning tool.



Caution Before cleaning any optical connector or fiber interface, ensure that all laser sources are switched off and that no optical radiation is present in the system.

6.2 Preventive Maintenance

The supplied FLIM LABS hardware components do not require routine internal maintenance.

Periodically inspect:

- External surfaces for mechanical damage
- Connectors and cables for wear, deformation, or contamination
- Optical connectors and fiber patch cables for cleanliness and damage
- Product labels for legibility

If any abnormal condition is detected, discontinue use of the affected component and contact technical support.

6.3 Service and Repairs

The supplied hardware components contain no user-serviceable internal parts. In the event of malfunction, accidental damage, or improper operation:

- Switch off and disconnect the affected component.
- Do not attempt internal repair.

- Contact the manufacturer's technical support for instructions.

Repairs or modifications must be carried out exclusively by FLIM LABS S.r.l. or authorized personnel. Use only original FLIM LABS spare parts and approved replacement accessories. Further warranty conditions are described in Section 1.6.



Unauthorized intervention or the use of non-original components may compromise safety, performance, and warranty coverage.

7 Decommissioning and Disposal

This chapter describes procedures for removing system components from service and disposing of them in accordance with applicable regulations.

7.1 Decommissioning Procedure

The supplied hardware components are designed for long-term laboratory use. When a component reaches the end of its technical or operational life, it must be decommissioned to prevent further use.

Decommissioning requires:

- Disconnecting the component from all power sources.
- Removing all signal, communication, and optical connections.
- Ensuring that the component cannot be unintentionally reused.

The same procedure applies in cases of:

- Permanent withdrawal from service
- Storage without intended future use
- Final disposal

Reuse of components after final decommissioning is not permitted unless explicitly authorized by the manufacturer.

7.2 Disposal

All hardware components supplied with the system must be disposed of in accordance with applicable local and European regulations governing electrical and electronic equipment.

In accordance with Directive 2012/19/EU (WEEE), electronic components must not be disposed of with household waste. The product labels include the WEEE symbol indicating the requirement for separate collection of electrical and electronic equipment. Contact your local waste management authority or a certified collection center for proper disposal procedures.

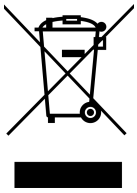


Figure 7.1: WEEE symbol